Ex: 7.

Flow Control out Data link layer using Sliding window protocol

Aim:

To write a program to implement flow control at the data link layer using the Sliding window protocol

Algorithm:

1) Start the program.

2) Read the total numbers of frames(N), window Size (10), the frame number to be lost.

3) Initialize:

base = 0 - first unacknowledged frow

nextSeq = 0 - next frame to be sent

4) While base < N, determine the Sender window from base to base + 10 - 1

5) Send all frames within the current window and increment nextSeq after sending each frame.

6) If there is no frame loss:

* Receiver accepts the frames in order

* Receiver Send an Ack for each received frame.

* Set base = nextSeq.

7) Repeat step 4-7 untill all frames are successfully transmitted & acknowleged.

8) Display that all frames.

9) Stop the program.

Output:

```
Enter total numbers of frame to Send: 3
Enter Sender Window Size: 2
Enter frames number to Simulate as lost
      (0 to 2, enter -1 for noilen): 1

Sender Window: From 0-Frame 1]
          Node A-> Sending Frame 0
          Node B-> Sending Frame 1
                (Frame 1 last in tranfinit)
          Node B-> Frame 1 missing
          Node A-> Timeout 1 Resending frame 1
Sender window: [Frame 1 - Frame 2]
          Node 4-> Sending Frame 1
          Node A -> Sending Frame 2
          Node B -> Recived Frame 1 Send Ack1
All 3 frame Succesfully transmitted from
      Node A to Node B.
```

Code:

```
Import Java. Util. Scanner:
   public class Siliding window{
     Static void Simulate Lint total frames,
       int window Size, Integer lost Frame){
       int base =0:
       int next seq=0;
       While (base < total frame)1
           int window End = Math. min (base +window
                                   Size -1, total Frame -1);
       System. out. println ("Selder window: [Frame ': base 1-
                 "Frame" + window End])
       if (last Fram != Null 88 next Seq == lost Frames)d
         System. out. print ("Frame ' + next Seq+ " Lost in
                                                  transut 1)"
            }next seq;
         3.
       if (lost Frame!= null 88 base ≤ lost Frame 82
                 lost frame Seq){.
         System. out. println ("node. B-> Frame "+lost
                       Frame +"missing");
         System. out. println ("Node A ->Timeout! Resending
           window Starting fram"+ lost Frame."."");
           next seq = lost Frame;
           lost Frame == null;
              Continue;
         for (int f=base; 6 < next Seq; f++){
           System. out. println ("node B-> Recived
           Frame"+ I+, sending Ack "+ 6): 3
```

```
        base = next Seq;);
      }.
public Static void main (String [] args){
        Scanner Sc = new Scanner (System.in);
    System.out.println ("Enter total number of frames to
    end ");
    Int total Frame = Sc.next Int();
    } else {
        lost Frame = input Last Frame; }.
        Simulate (total. Frame, window Size, lost Frame);
         Sc close (); }}.
```

Result:

Thus The Programe to implement flow Control at The Data Link layer Using The Slibding window protocal was implemented Succefdully.